

Poly-Metamath: Collaboration and collective reasoning in mathematics research

Introduction

Mathematics is a fundamentally social practice, yet most accounts of mathematical discovery are personal anecdotes from professionals or “lone genius” narratives. Further, the artifacts of research, mostly polished proofs and papers, rarely offer insight into the underlying reasoning process. One exception is *Polymath*, a series of web-based “projects” where anyone on the Internet help solve advanced and/or open mathematical problems by replying to other participants in blog comments [1]. The *Polymath* projects are not only a rich source of data for informal discussion and collaborative problem-solving, but also a model for successful knowledge generation outside of spatial, temporal, and professional bounds—*Polymath* contributors range in expertise from graduate students to Fields Medalists such as Timothy Gowers and Terence Tao.

Previous analyses of *Polymath* have used surface-level data, such as reply networks and response time dynamics, to identify properties of successful projects that could be generalized to other online platforms. In contrast, my work directly investigates what mathematicians are *thinking* during collaborative problem-solving. First, I characterize the dynamics of collective exploration and exploitation using a novel measure of “topic diversity” adapted from existing information-theoretic text analysis techniques [2]. I also develop a method of rating comments based on a taxonomy of cognitive “moves,” and discuss preliminary results.

Methods

Data collection and pre-processing. There are 20 total *Polymath* projects, with 16

involving original research and 4 “mini” projects for International Math Olympiad problems. I wrote a Python script to scrape comments and posts from all projects to create a corpus of 23,576 documents, including 15,924 comments. Using the HTML source code for the host blogs, I extracted and assigned 12 metadata fields for each document. An example comment is given in Table 1.¹

Diversity of mathematical topics. To quantify each comment’s topic structure, I first filtered for words containing meaningful mathematical content (e.g., terms for distinct sub-problems, strategies, or mathematical ideas). I then fit a probabilistic topic model using the MALLET package [3, 4], representing each comment as a distribution over 30 topics. Using Jensen-Shannon divergence (JSD) as a novel definition of topic diversity, I computed diversity in different project phases (i.e., chronological comment groups) based on the 30-topic distribution, as well as coarser 20-, 13-, and 8-topic distributions obtained by iteratively merging the most similar categories.

Cognitive category ratings. I helped define a taxonomy of 14 cognitive “moves” that appear in informal *Polymath* discussions by reviewing both empirical literature and first-person accounts of mathematical reasoning. I then wrote a script that uses OpenAI’s API to apply these definitions to each comment. Specifically, I created a Python class that prompts GPT-4o to assign “confidence ratings” for the categories to each comment. Example ratings are given in Table 2.

To test rating reliability, my collaborators and I separately rated a subset of 19 comments and compared this with two different runs of the GPT method. Human-human and GPT-GPT ratings had average

¹Data is available in a public repository: <https://bit.ly/polymath-data>

ID	Author	Time	In Reply To	# Equations
C-P5-Gr1478-5454	gowers	1264179060000.0	C-P5-G1478-5441	7

Original Comment
In this comment I want to try to clarify some muddled thoughts that are floating around in my head. If I fail, I may at least leave something in the mess that somebody else can beat into shape. [...] The next stage of the argument would be for bad guy to find such a set that intersected substantially with the set $\$A\$$ that good guy puts forward.

Table 1: An example *Polymath* comment with a subset of its metadata. Bolded terms are used to train a probabilistic topic model, which quantifies each comment as a distribution over 30 mathematically meaningful topics.

Deductive 0.5	Definitions 0.0	Strategic 1.0	Parallelism 0.0	Inductive 0.5	Speculative 0.7
Analogy 0.3	Relevance 0.3	Surprise 0.7	Impasse 0.6	Examples 0.6	Interpersonal 0.4

Table 2: Example confidence ratings from GPT-4o for the presence of 12 (of 14) cognitive “moves” in *Polymath* comment C-P5-Gr1478-5454. A rating of 1 means the comment fully embodies the cognitive category, while 0 means the comment does not have any characteristics of the category. Intermediate values reflect partial alignment or uncertainty in the rating.

correlations of 0.5420 and 0.5481, respectively, indicating that GPT ratings have the same variability as ratings by humans with mathematical and cognitive science domain knowledge.

Results

Diversity of mathematical topics in the first *Polymath* project is best characterized as a multi-scale phenomenon. Figure 1 shows that diversity decreases over time in the coarsest (8-topic) distribution, where topics align with general mathematical subfields. Conversely, Figure 2 shows that the relative diversity of the finest (30-topic) distribution increases over time, suggesting that contributors shift to exploring specific ideas within a subfield. Relative diversity reaches its maximum in the middle of the process, splitting the project into two

distinct phases: coarse-grained exploitation with fine-grained exploration, then overall exploitation in later stages. This suggests that successful collective problem-solving involves participants eventually converging upon a single approach, rather than each pursuing a specialized line of research over the entire duration of a project.

Cognitive moves (e.g., Figure 3) can clarify the nature of observed topic diversity. For instance, a high proportion of “example,” “analogy,” and “interpersonal” during a peak in topic diversity would suggest that participants are explicitly justifying their approach to others, rather than passively following one person’s work over time.

Future Directions

My next step will be to determine the

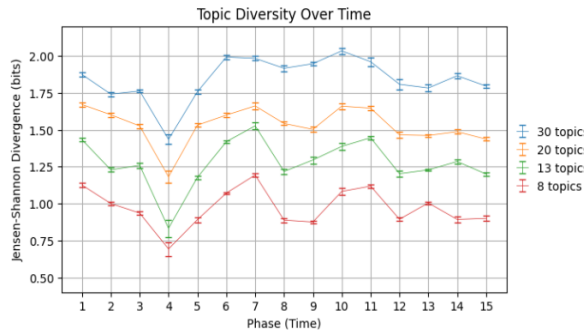


Figure 1: Absolute diversity of mathematical topics in the first *Polymath* project at different levels of granularity. Each phase contains an equal number of comments, ordered chronologically. Diversity of 8 coarse-grained topics decreases over time, indicating convergence on and exploitation of a particular mathematical subfield.

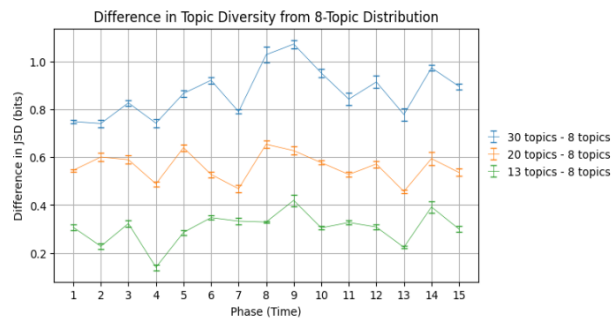


Figure 2: Diversity of mathematical topics in the first *Polymath* project relative to the coarsest (8-topic) distribution. The relative diversity of 30 fine-grained topics increases as contributors explore over a more compact space of ideas.

source of cognitive differences between projects. One approach is principal component analysis (PCA), which reduces the 14 categories into components that explain most of the variance between projects.

I hope other researchers will use the public *Polymath* dataset to characterize the cognitive basis of successful collaboration, as this work leaves many open questions. For example, do participants fill specific functional roles, such as only contributing empirical results, just as individual cognition is

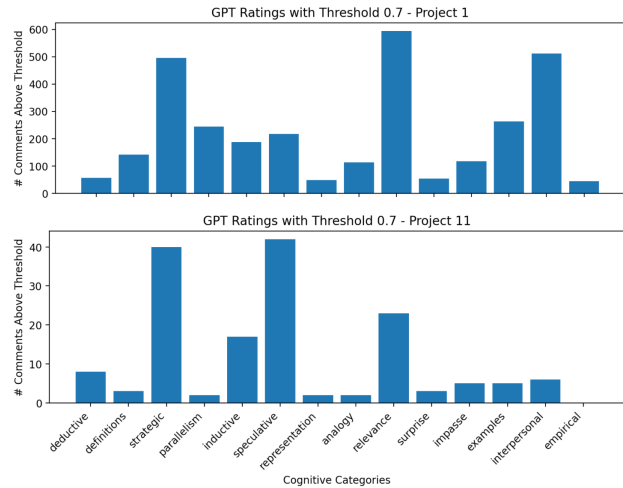


Figure 3: Number of comments that contain each cognitive “move,” determined by a threshold rating of 0.7. Each *Polymath* project has a distinct cognitive profile.

modularized?

I plan to apply insights from *Polymath* to my interests in developing computational models of learning and reasoning. In particular, understanding the what cognitive mechanisms underlie the communication of abstract ideas, such as the importance of analogy-making, will help me decide what mechanisms to implement in a theoretical model.

References

- [1] Gowers, T. et al. “Massively collaborative mathematics”. en. In: *Nature* 461.7266 (Oct. 2009), pp. 879–881.
- [2] Murdock, J. et al. *Exploration and Exploitation of Victorian Science in Darwin’s Reading Notebooks*. arXiv:1509.07175. Feb. 2017.
- [3] Blei, D. M. “Probabilistic topic models”. en. In: *Communications of the ACM* 55.4 (Apr. 2012), pp. 77–84.
- [4] McCallum, A. K. *MALLET: A Machine Learning for Language Toolkit*. 2002.